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Multi-scale autoregressive prediction method for structural seismic response integrating stationary wavelet transform and transformer

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ABSTRACT

Structural responses under seismic excitation are highly nonlinear, non-stationary and multi-scale. Conventional time-domain surrogate models are constrained in capturing high-frequency response details, and autoregressive sequence prediction frequently encounters exposure error issues, which subsequently lead to predicted time series drifts. This study proposes a structural response prediction framework that integrates stationary wavelet transform (SWT) with Transformer-based autoregressive prediction to address aforementioned challenges. SWT is employed to decompose ground motions and structural responses into multi-scale coefficients, which enables effective separation and characterization of low-frequency trends and high-frequency details of signals. A Transformer-based encoder-decoder model is then constructed to autoregressively predict future response coefficient sequences within the coefficient domain, followed by the restoration of time-domain predictions through inverse transform reconstruction. To mitigate exposure errors in the autoregressive progress, channel-wise noise injection and self-conditioning training strategies are introduced during the training phase, thereby enhancing the model's stability. Numerical test results demonstrate that the proposed method outperforms traditional time-domain models in terms of overall prediction accuracy, exhibiting better performance in high-frequency response characterization and error suppression. Sensitivity analyses of key hyperparameters reveal the governing laws of different parameter configurations on model performance. Ablation studies further validate the necessity of multi-scale representation in improving both the prediction accuracy and the stability of the model.

1. Introduction

The rapid acquisition and prediction of key structural dynamic responses under seismic excitations constitute a technical foundation for rapid post-earthquake assessment [1–3]. Achieving real-time and post-event response evaluation directly enables the operation of early warning, condition assessment [4]. Traditional non-linear time history analysis is computationally impractical for engineering applications, necessitating the development of efficient surrogate models for rapid inference [5–7]. In recent years, data-driven models, such as convolutional neural networks (CNNs) [8,9], long short-term memory network [10–12] and attention-based Transformers [13–15] have demonstrated significant potential in time-series prediction [16]. Compared to complex finite element (FE) analysis and parameter identification [17–19], data-driven models can directly learn the mapping relationships between ground motions and structural responses from historical observations, offering advantages in modeling flexibility and end-to-end deployment [20].

However, seismic excitations and structural responses generally exhibit strong non-stationary characteristics [21], which include both slowly evolving low-frequency trends and transient high-frequency peaks [22]. Low-frequency components are dominated by structural modes, characterizing the processes of global deformation and energy evolution, whereas high-frequency components are associated with material nonlinearity and damage evolution [23]. Time-domain models must learn these different components simultaneously from the same response sequence. Because low-frequency components generally contain more energy and have larger numerical amplitudes, the optimization process may be dominated by the global response trend, while short-duration high-frequency peaks may be smoothed or exhibit phase distortion. This scale imbalance is an important reason why direct time-domain models can reproduce the overall waveform but struggle to

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preserve local response details [24].

Wavelet transform can effectively characterize non-stationary signal features in time-frequency domain [25], which facilitates signal's multi-scale domain representation [26]. Recent studies have demonstrated that wavelet-based neural networks can exploit multi-scale representations for time-series and seismic-response modeling [27–29]. In existing studies, Conventional discrete wavelet transform (DWT) is generally used for denoising, feature extraction, dimensionality reduction or feature enhancement [30]. The resulting features, including either preprocessed time-domain signals or wavelet-domain coefficients, are then supplied to CNN, LSTM or Transformer models for prediction [31]. However, DWT involves downsampling operations, which makes temporal alignment across different scales difficult [32]. Consequently, coefficients from different frequency bands are often treated as independent feature streams or predicted separately before being fused to reconstruct the time-domain response. This procedure may weaken the temporal correspondence among different scales during prediction. The stationary wavelet transform (SWT) eliminates downsampling and preserves one coefficient at every original time index [33], allowing approximation and detail coefficients to be jointly modeled at the same temporal positions. This property makes SWT suitable for combining multi-scale signal representation with autoregressive sequence prediction while preserving the temporal correspondence required for recursive modeling. In addition, the temporally aligned wavelet-coefficient channels provide cross-scale constraints, which may improve the consistency and stability of the reconstructed response.

In real-time structural response prediction, the traditional autoregressive algorithm generates future sequences through a recursive mechanism [34], which inevitably causes error accumulation that leads to amplitude and phase distortions. This recursive setting introduces a train-inference inconsistency, referred to here as exposure error. During training, the model is conditioned primarily on ground-truth historical responses, whereas during rolling inference it is conditioned on its own previously generated responses. Consequently, the conditional distribution encountered during inference differs from that used during optimization. Once a prediction deviates from the ground-truth response, this deviation is fed back into the next prediction step and changes the subsequent conditioning state. The model may therefore encounter response histories that are poorly represented in the training data, causing local errors to propagate and amplify throughout the recursive process. Exposure error thus refers to the distribution-shift mechanism caused by the mismatch between ground-truth and self-generated histories. The resulting distribution shift can cause local prediction errors to propagate into long-sequence drift. To mitigate these issues, the Sequence-to-Sequence (Seq2Seq) modeling framework has been proposed [35–37]. By leveraging an encoder-decoder architecture, this framework directly predicts a segment of future sequence given past observations, with the model optimization being guided by multi-step sequence-level supervisory signals [38]. Compared to single-step autoregression, Seq2Seq models can reduce local error propagation [39], while these errors may still amplify during the iterative process when integrated with sliding windows in rolling prediction scenarios long-sequence [40,41]. To address this problem, strategies such as noise injection [42] and scheduled sampling [43,44] have been widely adopted to improve model robustness. These methodologies are effective for enhancing the stability of long-sequence autoregressive predictions.

Based on aforementioned limitations, this study proposes a structural response prediction framework that integrates SWT-based multi-scale representation with Transformer-based autoregressive prediction to improve the prediction accuracy and generalization ability. Unlike existing time-domain CNN, LSTM and Transformer models, the proposed framework does not directly predict the structural response as a single raw time-domain sequence. Instead, SWT is first used to decompose the seismic excitation and structural response into temporally aligned multi-channel wavelet-coefficient representations. A Transformer encoder-decoder then autoregressively generates future response coefficients in the coefficient domain, and the time-domain response is reconstructed through the inverse SWT. Furthermore, to enhance the stability of long-sequence rolling predictions, noise injection and self-conditioning training strategies are introduced to strengthen the robustness of the model against error propagation. Finally, the stability of the proposed framework and the necessity of the key components are validated through comprehensive parameter analyses and ablation studies.

2. Structural seismic response prediction framework

2.1. Overall architecture

The proposed structural seismic response prediction framework is illustrated in Fig. 1. This framework comprises three modules, namely the multi-scale feature processing module, the Transformer autoregressive prediction module and the signal reconstruction module:

(1) Multi-scale feature processing module: The module utilizes the SWT to perform multi-scale decomposition on seismic excitations and structural responses. Subsequently, input features are constructed via a temporal offset concatenation strategy to simulate the excitation and state terms required for the recursive solution of structural dynamics.
(2) Transformer Autoregressive prediction module: The module uses a Transformer encoder-decoder to approximate the recursive evolution of structural dynamic response. The encoder is responsible for extracting latent variable representations related to structural damage states, while the decoder generates serialized predictions of multi-scale response coefficients in an autoregressive way.
(3) Signal reconstruction module: The module concatenates the output multi-scale coefficients and performs reversible reconstruction through the inverse SWT (ISWT). This process recovers the time-domain structural response sequence, achieving a mapping from coefficient domain to time domain.

2.2. Multi-scale feature processing module

2.2.1. Initial setting

The recursive prediction mechanism of autoregressive models is highly dependent upon the initial damage state provided by the encoder, which is determined by the observation window of length L_{enc}. However, at the onset of seismic excitation ($t = 0$), the system lacks prior information [1]. To address this, a consistent zero-padding strategy is adopted during training and inference phases [45].A zero-value sequence of length L_{enc} is prepended along the temporal axis before the starting point of the ground motion input sequence E and its corresponding structural response sequence R, as illustrated in Fig. 2. During model training, zero padding is introduced to characterize the initial undamaged structural features. During inference, this strategy is utilized to construct an initial encoding window, thereby ensuring the initiation of autoregressive predictions. Zero-padding should be executed prior to the multi-scale decomposition of signals to maintain the continuity of temporal features and to prevent the degradation of time-domain statistical characteristics.

2.2.2. Multi-scale feature representation

Given the non-stationary characteristics of seismic excitations and structural responses, SWT is utilized to decompose the signal into multi-scale characteristic. Compared to the DWT, SWT possesses shift-invariance, ensuring that features across various scales achieve strict alignment in the temporal dimension. Through N-level decomposition, the original one-dimensional L-length time series $z_t \in \mathbb{R}^{L\times 1}$ is trans-

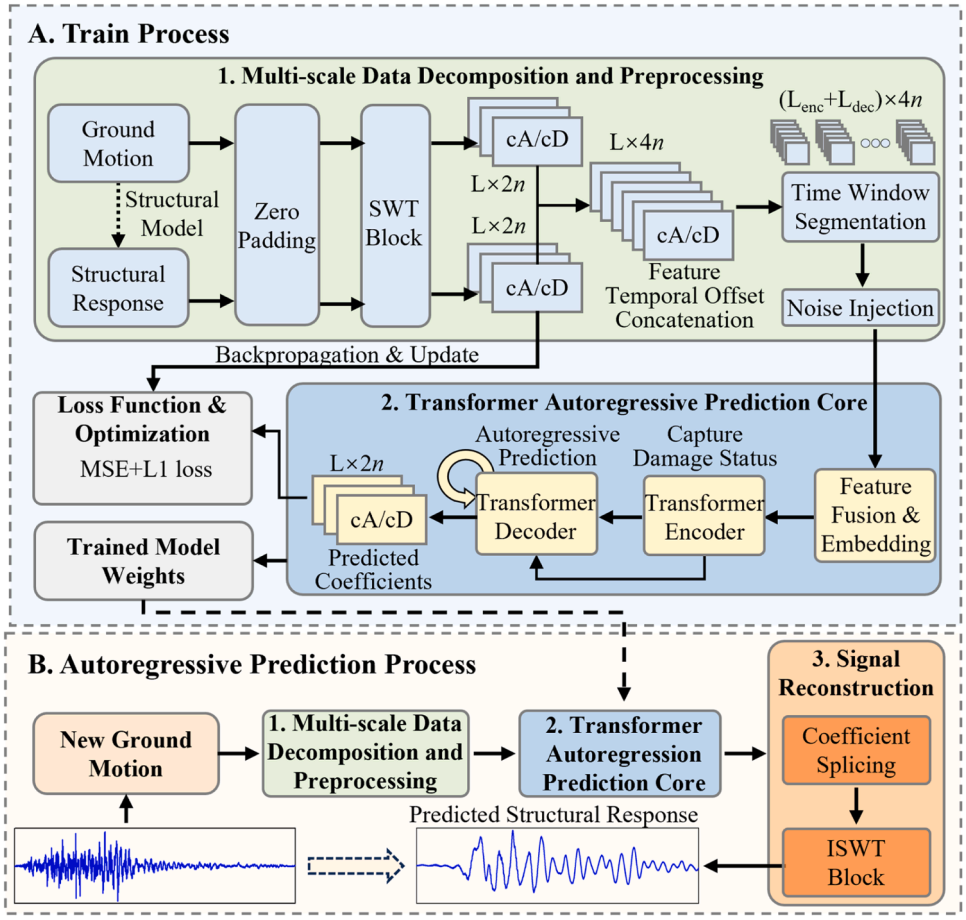


Fig. 1. Overall architecture of the proposed autoregressive framework for structural seismic response prediction integrating SWT and Transformer.

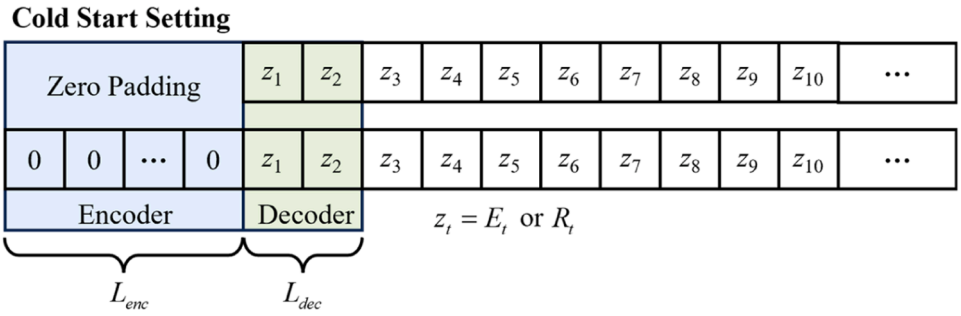


Fig. 2. Schematic representation of the zero-padding strategy for the start problem.

formed into a multi-scale space $\mathbb{R}^{L\times 2N}$ consisting of approximation coefficients $cA_j(t)$ and detail coefficients $cD_j(t)$:

$$z(t)\xrightarrow{SWT}\{cA_N(t), cD_N(t)\cdots cA_1(t), cD_1(t)\} \tag{1}$$

where $cA_j(t)$ characterize the low-frequency trends of the signal, $cD_j(t)$ captures high-frequency local abrupt information, and $j\in\{0,1,\ldots,N-1\}$ denotes the decomposition level. Figure A1 in Appendix A presents an example of the SWT decomposition for a typical function.

The coefficients cA_{j+1} and cD_{j+1} at the $(j+1)^{\text{th}}$ level can be obtained recursively from the j-th level coefficient cA_j via convolution operations, which reflects the significant structural redundancy of SWT:

$$cA_{j+1}[k]=\sum_l h_j[l]cA_j[k-l] \tag{2}$$

$$cD_{j+1}[k]=\sum_l g_j[l]cA_j[k-l] \tag{3}$$

where k is the discrete time index, l represents the filter tap index, and h_j and g_j denote the low-pass and high-pass filters at the j-th level, respectively. To guarantee the mathematical completeness of the decomposition process and reduce boundary effect, the signal length L must satisfy the 2^N divisibility constraint, and the number of decomposition levels N must satisfy the theoretical limit of $N\le|\log_2(L)|$ [33].

2.2.3. Feature fusion and data augmentation

The temporal evolution of structural seismic responses retains three critical elements, namely the past state of the structure $S_{0:t-1}$, the response of the previous R_{t-1} and the current seismic excitation E_t. Referring to the structural dynamic equilibrium equation, past state is

determined by the path-dependent nature of the damping and stiffness matrices [46], where parameters evolve according to the cumulative damage and nonlinear behavior. Given that damage states are typically unobservable, this study adopts a past response sequence of length L_{enc} as a state proxy variable to indirectly characterize the structural damage accumulation information. To establish state-transition mappings among these variables, a temporal offset concatenation strategy is introduced to align and combine R_{t-1} and E_t along the temporal axis (Fig. 3) to simulates the recursive updates based on the existing state and external excitations.

Compared to the conventional full-sequence mapping mode, the sliding window segmentation mechanism employed in this study (Fig. 3) effectively resolves the confliction between data scale and model representation capacity in time series modeling [45]. During the sliding window segmentation process, the Encoder segment represents the known part of the signal, while the Decoder segment is typically considered as the unknown part. The stride step only affects the number of segments after the segmentation of ground motion samples for training or prediction, but does not affect the prediction length of each Encoder-Decoder segment, that is, the decoder part. This mechanism not only reduces the number of model parameters, but also achieves efficient sample expansion through window sliding and overlapping. This improves data utilization and reduces the risk of overfitting. Furthermore, sliding windows allow the model to focus on local dynamic features [47], which is more consistent with the short-term evolution laws of nonlinear dynamic responses. Sample expansion must be executed after the multi-scale feature representation, to avoid the distortions generated at window edges during the scale decomposition process (Figure A1 in Appendix A). This prevents local samples from deviating from the global statistical distribution due to boundary truncation effects.

2.2.4. Noise injection

In long-sequence autoregressive prediction, exposure error is a critical factor affecting the model accuracy and stability. From the perspective of dynamic evolution, this recursive inference process can be considered as the iterative integral update of response components over time. Generally, integral operators possess an inherent smoothing effect on high-frequency disturbances [48], while it has a cumulative amplification effect on low-frequency biases [49]. From frequency-domain perspective, exposure error manifests as minute offsets in the low-frequency components of the structural response, which subsequently evolve into global drift terms through continuous integration during the rolling prediction process [50].

Therefore, a noise injection strategy is introduced during the training phase. For the j-th SWT channel, the normalized coefficient $z_{j,t}$ is defined as:

$$z_{j,t} = \frac{c_{j,t} - \mu_j}{\sigma_j} \tag{4}$$

where $c_{j,t}$ denotes the SWT coefficient in its original physical unit, and μ_j and σ_j are the mean and standard deviation of the j-th channel calculated from the training set. Gaussian noise $\epsilon_{j,t}$ is then added to the normalized coefficient according to:

$$\tilde{z}_{j,t} = z_{j,t} + \alpha\varepsilon_{j,t},\ \varepsilon_{j,t} \sim \mathcal{N}(0,1) \tag{5}$$

where α denotes the noise standard deviation in the normalized coefficient space. After transforming the perturbation back to the physical coefficient space, the perturbation of the j-th SWT channel $\Delta c_{j,t}$ can be expressed as:

$$\Delta c_{j,t} = \alpha\sigma_j\varepsilon_{j,t} \tag{6}$$

Consequently, the actual perturbation amplitude in physical units depends on the standard deviation of each SWT channel. The corresponding time-domain perturbation is Δx_t obtained by applying the inverse SWT to the perturbed coefficient set:

$$\Delta x_t = ISWT\left(\{\Delta c_{j,t}\}_{j=1}^{J}\right) \tag{7}$$

Because SWT is redundant and the coefficient channels at different scales are not statistically independent, the root-mean-square (RMS) perturbation of the reconstructed time-domain signal is generally not equal to $\boldsymbol{\alpha}$ times the amplitude of the original physical signal. The noise is applied with the same normalized intensity $\boldsymbol{\alpha}$ across all channels. However, this does not imply the same physical disturbance amplitude across channels, because the corresponding physical perturbation standard deviation is $\boldsymbol{\alpha\sigma_j}$. In this way, each channel receives the same relative perturbation with respect to its own training-set variability, while the actual disturbance in the original physical units remains channel-dependent. Due to the smoothing characteristics of the recursive prediction process, disturbances in high-frequency channels are generally attenuated more rapidly, whereas small low-frequency biases may accumulate over successive prediction steps. The noise-injection

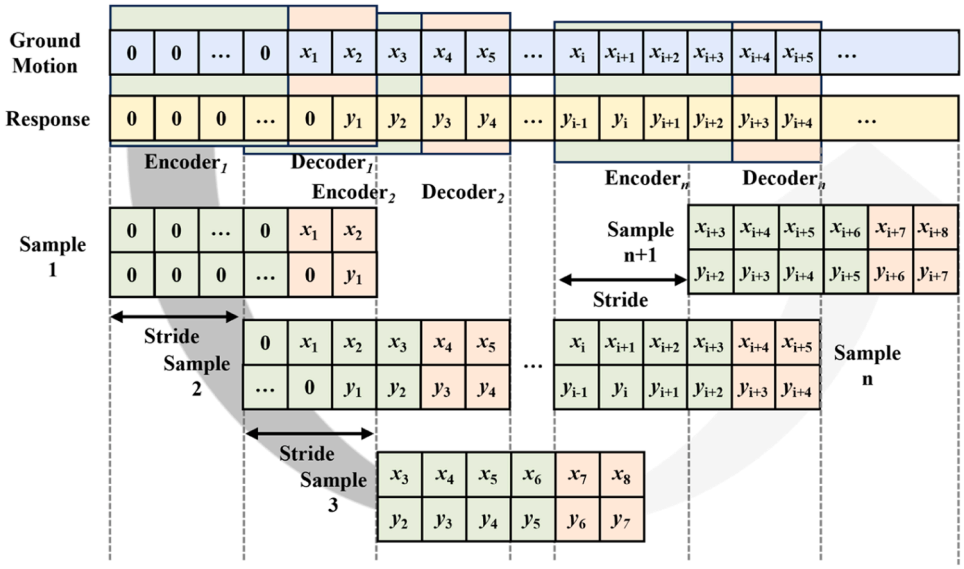


Fig. 3. Feature temporal offset concatenation considering structural dynamic recursive relationships and data augmentation by sliding window techniques.

strategy therefore exposes the model to non-ideal coefficient conditions while preserving the essential high-frequency transient information.

2.3. Transformer-based autoregressive prediction module

This section constructs a deep learning model based on a Transformer encoder-decoder [51], aiming to establish a complex nonlinear mapping between the coefficient domain of seismic excitations and structural responses, as given in Fig. 4. The encoder, with a window length of L_{enc}, is designed to extract latent feature representations of the structural evolution state. Under the constraints of latent states, the decoder, with a window length of L_{dec}, autoregressively generates predicted responses by incorporating future excitation sequences. Both the inputs and outputs of this module are defined within the multi-scale coefficient space, and the final prediction results are reconstructed into the time domain via the ISWT described in Section 2.4.

To enhance the robustness of predictions and suppress the distribution shift between the training and inference phases, a self-conditioning strategy is introduced into the Seq2Seq framework. During the decoder recursion process, the predicted output of the model itself is used as the input condition for the subsequent step. This paradigm can be viewed as an evolution of scheduled sampling under an extreme probability distribution, aimed at maintaining a high degree of consistency between the closed-loop feedback logic in the training phase and that in the inference phase [43]. This autoregressive modeling paradigm essentially simulates the recursive evolution process of a dynamic system, and its specific prediction procedure is as follows:

(1) Construction of Encoder/Decoder Segments: Suppose at time t, the seismic excitation and structural response coefficient sequences in the multi-scale domain are E and R, respectively. The encoder input is defined as $X = [E_{t-L_{enc}+1:t}, \tilde{R}_{t-L_{enc}:t-1}] \in \mathbb{R}^{L_{enc} \times 4N}$. $\tilde{R}$ represents the historical response sequence, which takes the ground-truth values during the training phase and the previously generated predictions during the inference phase. The decoder progressively generates $\hat{R}_{t+1:t+L_{dec}}$ within the prediction window. At the i-th recursive step ($i \in [1, L_{dec}]$), the conditional input vector for the decoder is $Y^{in}_{t+i} = [E_{t+1:t+i}, \hat{R}_{t:t+i-1}]$, and the output vector is $Y^{out} = \hat{R}_{t+1:t+i}$. Within a single prediction window of length L_{dec}, the encoder input X and its output representation remain constant. Upon entering the next window, a new X is constructed and re-encoded.

(2) Representation of Structural Damage State: By feeding X into the encoder, latent variable sequence H, which reflect the global structural state, are extracted through the self-attention mechanism. The terminal state representation $h_t = [E_t, \tilde{R}_{t-1}]$ is the conditional information for the decoder to constrain the generation of future responses.

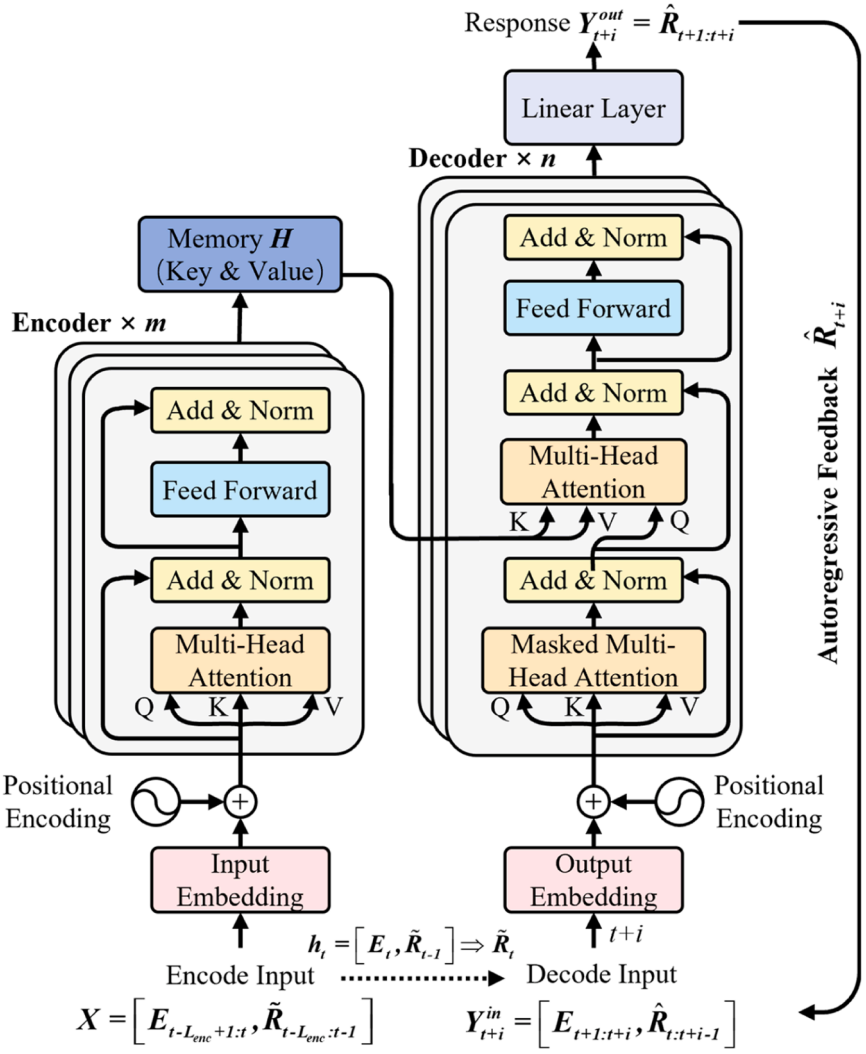


Fig. 4. Detailed architecture of the Transformer-based encoder-decoder module for self-conditional autoregressive prediction.

(3) Self-conditioned State Evolution Recursion: The initial response condition $\widehat{\boldsymbol{R}}_{\boldsymbol{t}}$ for the decoder is constructed based on the encoder output representations ($\boldsymbol{H}$ and $\boldsymbol{h}_{\boldsymbol{t}}$). For the $\boldsymbol{i}^{\text{th}}$ prediction step ($\boldsymbol{i} \in [\boldsymbol{1}, \boldsymbol{L}_{\mathbf{dec}}]$), the decoder calculates the response prediction for the current step, $\widehat{\boldsymbol{R}}_{\boldsymbol{t+i}} = \text{Decoder}(\boldsymbol{E}_{\boldsymbol{t+1:t+i}}, \widehat{\boldsymbol{R}}_{\boldsymbol{t:t+i-1}}, \boldsymbol{H})$, under the joint conditions of the generated response sequence $\widehat{\boldsymbol{R}}_{\boldsymbol{t:t+i-1}}$ and the corresponding future excitation $\boldsymbol{E}_{\boldsymbol{t+i}}$ through masked self-attention and cross-attention mechanisms.

(4) Closed-loop Feedback Update Mechanism: The predicted response $\widehat{\boldsymbol{R}}_{\boldsymbol{t+i}}$ generated at the current step is fed back to the decoder input, which is combined with the excitation of the next step $\boldsymbol{E}_{\boldsymbol{t+i+1}}$ to construct the subsequent conditional input $\boldsymbol{Y}^{\boldsymbol{in}}_{\boldsymbol{t+i+1}}$, thereby driving further predictions, as shown in Fig. 5. This closed-loop feedback mechanism embodies self-conditioning recursion, ensuring that the decoder consistently utilizes its own predictions as the response condition for the next step.

(5) Termination and Output: Upon completion of the $\boldsymbol{L}_{\mathbf{dec}}$ recursive steps, the autoregressive generation process terminates, and the multi-scale response coefficient sequence $\widehat{\boldsymbol{R}}_{\boldsymbol{t+1:t+L}_{\mathbf{dec}}}$ within the prediction window is output. Subsequently, $\widehat{\boldsymbol{R}}$ is input into the ISWT for reconstruction to obtain the corresponding time-domain structural response prediction sequence.

2.4. Signal reconstruction module

The output of the Transformer autoregressive prediction module consists of the predicted coefficients of structural responses within the multi-scale feature space. To obtain the intuitive structural response, the predicted coefficients are mapped back to the time domain through an inverse transform. Because the sliding window partition and the autoregressive recursive mechanism result in output sequences with piecewise characteristics (as illustrated in Step 1 of Fig. 6), boundary high-frequency oscillation effects [52] can be induced if ISWT are performed directly on each sub-segment followed by time-domain stitching. Such oscillations compromise the physical rationality and spectral consistency of the reconstructed signals.

Consequently, the strategy of coefficient-domain integration followed by global time-domain reconstruction is adopted to restore the response signals. The predicted coefficients generated from each sliding window are concatenated sequentially along the temporal axis (Step 2 of Fig. 6). Subsequently, a unified ISWT operation is performed on the concatenated complete coefficient matrix (Step 3 of Fig. 6) to achieve a reconstruction of the structural response. This strategy effectively suppresses the boundary effects induced by window truncation and enhances the smoothness and numerical stability of the reconstructed signals.

2.5. Loss function

Structural seismic response signals exhibit non-uniform distribution in the frequency domain. Given the significant differences in energy magnitudes across different frequency bands, this study introduces scale-specific normalization, where Z-score normalization is performed independently for each sub-band channel within the multi-scale space. This strategy is designed to eliminate the numerical masking effect of low-frequency coefficients on high-frequency details, enabling the equal weighted perception of cross-band features.

In terms of optimization criteria, a joint loss function consisting of Mean Squared Error (MSE) and Mean Absolute Error (MAE) is formulated:

$$\begin{aligned} L_{\text{total}}(\boldsymbol{y}, \widehat{\boldsymbol{y}}) = & \underbrace{\frac{1}{\boldsymbol{B} \times \boldsymbol{T} \times \boldsymbol{C}} \sum_{b=1}^{B} \sum_{t=1}^{T} \sum_{c=1}^{C} \left\| \boldsymbol{y}_{b,t,c} - \widehat{\boldsymbol{y}}_{b,t,c} \right\|_2^2}_{L_{\text{MSE}}} \\ & + \underbrace{\frac{1}{\boldsymbol{B} \times \boldsymbol{T} \times \boldsymbol{C}} \sum_{b=1}^{B} \sum_{t=1}^{T} \sum_{c=1}^{C} \left\| \boldsymbol{y}_{b,t,c} - \widehat{\boldsymbol{y}}_{b,t,c} \right\|_1}_{L_{\text{MAE}}} \end{aligned} \tag{8}$$

where $\boldsymbol{B}$ denotes the batch size, $\boldsymbol{T}$ is the sequence length (consistent with $\boldsymbol{L}_{\mathbf{dec}}$), $\boldsymbol{C}$ is the number of output response-coefficient channels in the multi-scale domain, and $\boldsymbol{y}$ and $\widehat{\boldsymbol{y}}$ represent the target and predicted values of the multi-scale coefficients, respectively. The normalization factor in the loss is therefore $\boldsymbol{B} \times \boldsymbol{T} \times \boldsymbol{C}$, which represents the total number of predicted coefficient elements involved in the mean error calculation. By penalizing large error terms quadratically, MSE accelerates convergence, enabling the model to quickly capture trends and fit extreme amplitudes. In contrast, MAE employs a linear penalty mechanism, effectively mitigating the effects of anomalous disturbances, thereby maintaining waveform phase consistency, and significantly enhancing the robustness of the model [53].

3. Numerical test

3.1. Dataset construction

To verify the effectiveness and generalization ability of the proposed framework, a numerical test is conducted. Specifically, a structural FE model and ground motion database are established, and paired seismic excitation and structural response data are further obtained by nonlinear time history analysis. Furthermore, preprocessing procedures are performed to establish a standardized dataset for model training, validation and testing.

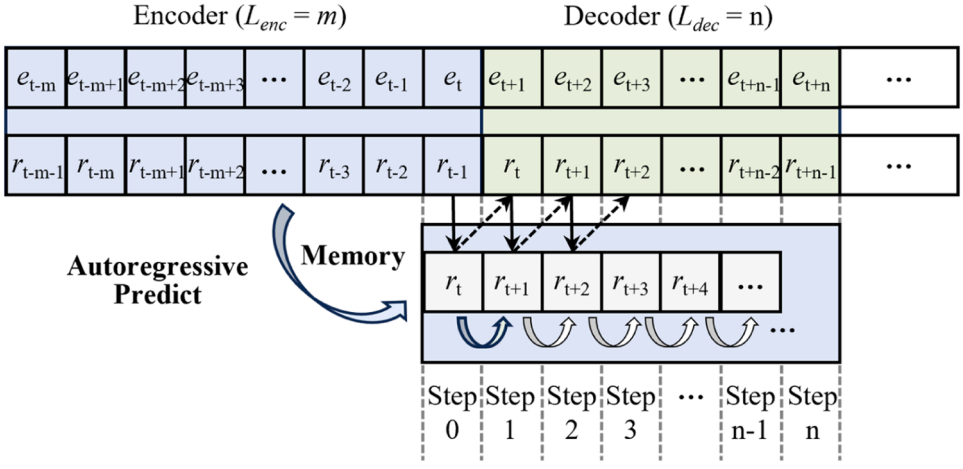


Fig. 5. Schematic representation of the sequential prediction and feedback update logic for self-conditional autoregression.

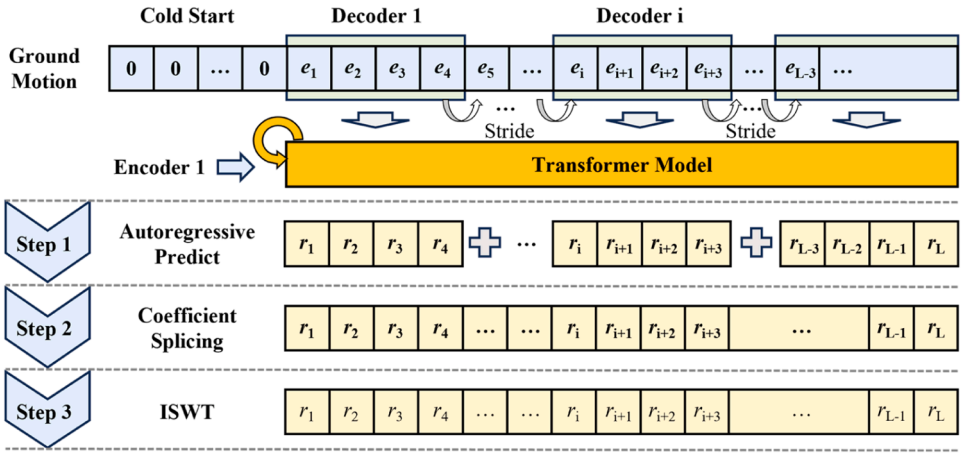


Fig. 6. Signal reconstruction procedure involving coefficient-domain concatenation and ISWT.

3.1.1. Benchmark structural and finite element model

A 4 × 40 m continuous beam bridge is selected with solid rectangular single-column piers (Fig. 7). The beam is composed of concrete with compressive strength of 50 MPa (C50), while C35 is used for the piers and C30 for the foundations and piles. Hot-rolled ribbed bars (HRB400) are employed for the longitudinal bars in the piers. The total weight of the superstructure is 25,090 kN. Lead-rubber bearings (LRBs) with dimensions of 920 × 920 mm are utilized, featuring a yield force of 417 kN and a rubber shear modulus of 1 GPa. Two 1.5 m diameter piles with a length of 43.5 m are set under each foundation. The dynamic coefficient of the site soil is 7000 kN/m^4, and the vertical support capacity of a single pile is 8000 kN.

A nonlinear FE model of the test bridge is established using the OpenSees, as shown in Fig. 8. To capture the nonlinear dynamic behavior of piers under strong earthquakes, displacement-based fiber beam-column elements with fiber sections are used for modeling. Structure-soil interaction is simplified using a six-spring model. The seismic isolation bearings are simulated using a bilinear mechanical model, and Rayleigh damping is adopted with a modal damping ratio of 0.05.

3.1.2. Ground Motion Dataset

To enhance the model's generalization capability across various seismic excitations, 160 typical ground motion records selected by Baker et al. [54] from the PEER ground motion database are utilized. The detained information of these ground motion records can be found in https://www.jackwbaker.com/PEER_gms.html and are summarized in Tables A1-A4 of Appendix A. This dataset includes four distinct categories of earthquake characteristics (Fig. 9). Set #1a consists of 40 broadband ground motions selected to match the target response spectrum for a magnitude 7 strike-slip earthquake at a distance of 10 km on a soil site. Set #1b comprises 40 broadband ground motions selected using the same procedure but targeting a magnitude 6 strike-slip earthquake at a distance of 25 km on a soil site. Set #2 consists of 40 broadband ground motions for a magnitude 7 strike-slip earthquake at a distance of 10 km on a rock site. Set #3 is the pulse-like near-fault ground-motion set, consisting of 40 unscaled three-component records containing strong velocity pulses of varying periods in their fault-normal components. By incorporating diverse distributions of magnitudes, epicentral distances and site conditions, it effectively captures a broad spectrum of earthquake scenarios.

3.1.3. Training dataset and preprocessing

Given the inconsistencies in the duration and sampling frequency of the original records, cubic spline interpolation [55] is employed for resampling to standardize all ground motion records into equal-length sequences with a duration of 50 s and a sampling frequency of 100 Hz. This interpolation changes the sampling interval but does not by itself unify the record durations. Therefore, a fixed duration of 50 s, corresponding to 5000 samples, is subsequently imposed. Records longer than 50 s are truncated to the first 50 s, whereas records shorter than 50 s are zero-padded at the end to reach 50 s. The zero-padding represents the absence of prescribed ground motion after the end of a short record and does not introduce additional measured data. The

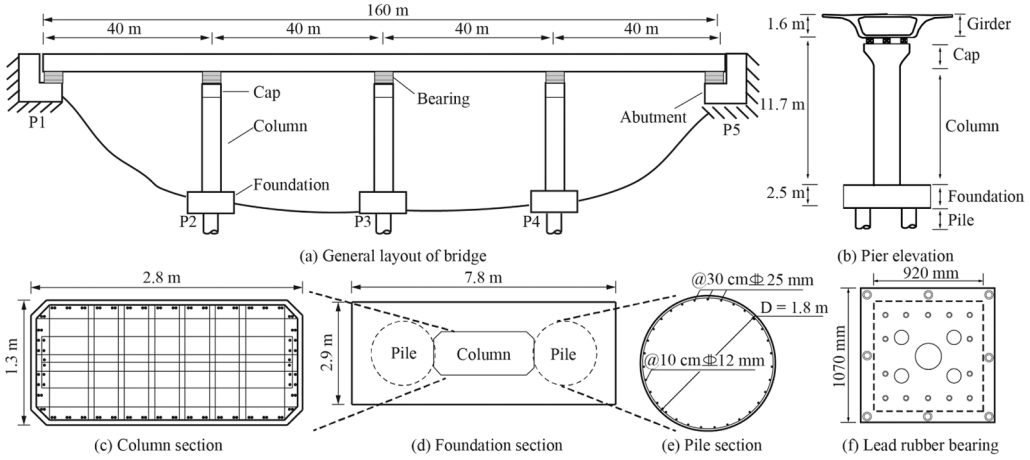


Fig. 7. Structural layout and configuration of the prototype bridge.

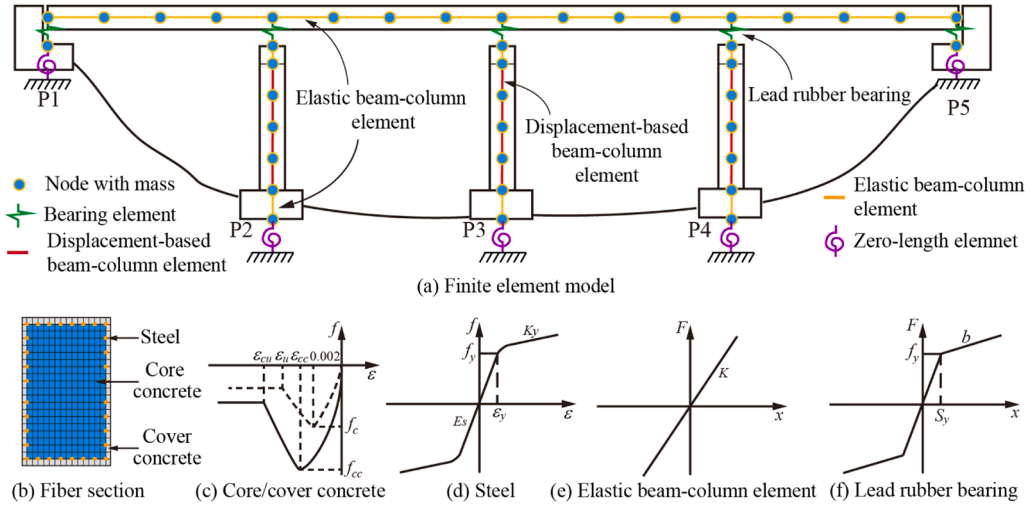


Fig. 8. Finite element model of the prototype bridge.

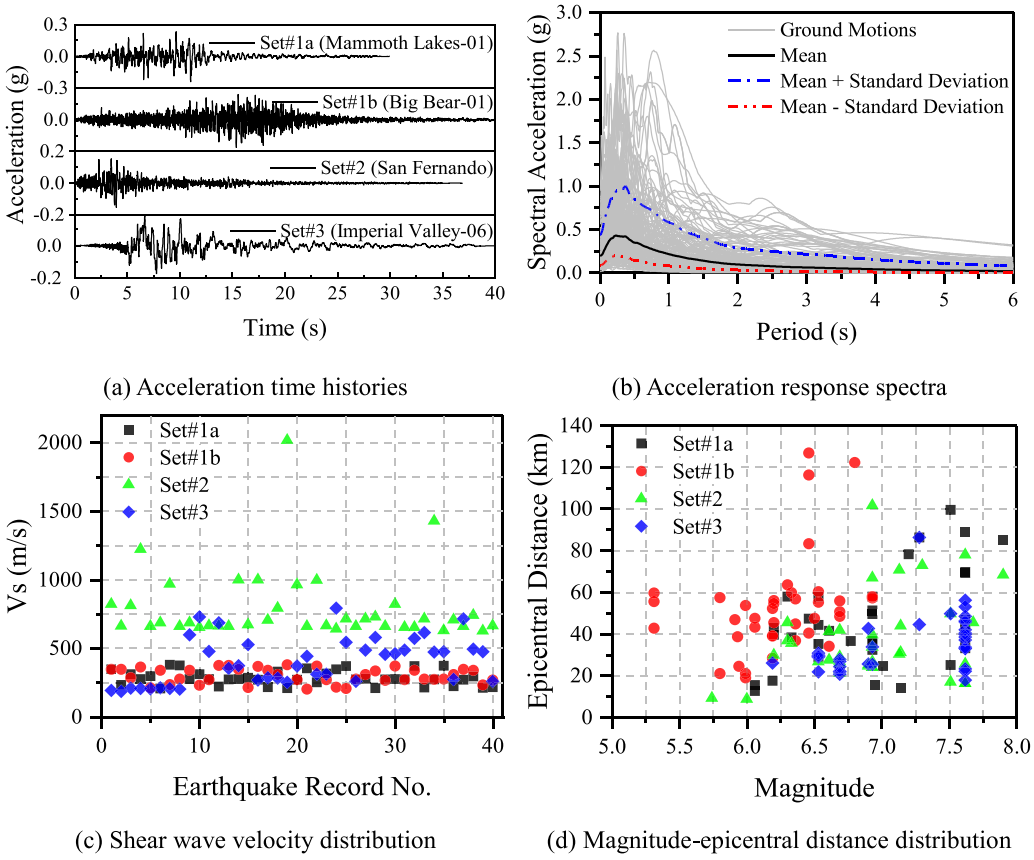


Fig. 9. Time-frequency characteristics and statistical distributions of the ground motion dataset.

resulting standardized signals are input into the FE model for nonlinear time-history analysis, and the bending moment at the base of pier P3 is extracted as the representative output over the same 50 s analysis window. This fixed-duration design ensures that each sample has an equal weight, thereby ensuring sample balance within the training set. The resampled signals are input into the FE model for nonlinear time history analysis, and the bending moment at the base of pier P3 is extracted as the representative output. Note that, owing to the considerable variation in seismic intensity, the resulting responses include yielding and nonlinear hysteretic behavior of pier P3 and the bearings under strong earthquake scenarios; however, the dataset does not systematically cover local failure or collapse-level damage. To prevent data leakage, the seismic records are partitioned pseudo-randomly (with a random seed of 42) into training, validation and test sets according to a 7:2:1 ratio, containing 112, 32 and 16 independent sequences, respectively. The partitioning was performed at the level of complete seismic records before SWT decomposition, normalization and sliding-window generation; therefore, no record was shared across different subsets. Subsequently, the ground-motion inputs and corresponding structural responses were decomposed using SWT. For each SWT channel, the

mean and standard deviation were calculated exclusively from the complete SWT sequences in the training set. These training-set statistics were then fixed and applied to the validation and test sets without recalculation. Channel-wise normalization was performed before sliding-window segmentation, and the same training-set statistics were used for inverse normalization after prediction.

In the feature representation phase, a 5-level SWT based on the Daubechies 4 (db4) wavelet base is performed on the zero-padded excitation and response sequences, which yields a 20-dimensional multi-channel feature space (10 excitation and 10 response channels). A sliding window is applied for data augmentation, with the encoder length L_{enc}, decoder length L_{dec} and stride step configured to 128, 64 and 8, respectively. Therefore, the final database contains 68,598 training samples, 19,776 validation samples and 9888 test samples. Note that the stride of 8 is used only for constructing overlapping training windows. Although adjacent training samples share part of their encoder and decoder ranges, each sample is used independently for supervised learning, and their outputs are never concatenated. During inference, the window start is determined by the decoder length (64 samples). Consequently, adjacent decoder output intervals are contiguous and non-overlapping. The valid predictions are concatenated in chronological order without averaging or any other overlap-resolution operation.

The autoregressive architecture is implemented using the PyTorch framework on a hardware platform equipped with an Intel® Xeon® Gold 6248 CPU and an NVIDIA RTX 3090 GPU. The model is trained for 100 epochs, with each epoch consisting of 134 mini-batches and a batch size of 512. The AdamW optimizer is selected with an initial learning rate of 1×10^{-4}.

3.2. Performance evaluation metrics

To evaluate the model performance, Coefficient of determination (R^2) is selected as the core indicator for evaluating the goodness of fit due to its intuitiveness [45]:

$$R^2 = 1 - \frac{\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{m}(y_i - \bar{y})^2} \tag{9}$$

where m represents the total number of sampling points, $\bar{y}$ is the average value of the target responses, and y_i and $\hat{y}_i$ denote the target and predicted values of the i-th point, respectively.

Given that the damage evolution of engineering structures is closely related to peak responses, the ability of model to capture extreme values directly affects the accuracy of post-disaster damage assessment and safety determination [56]. Peak Percentage Error (PPE) is thus introduced as a specialized evaluation metric to quantitatively characterize the prediction deviation of the model regarding peak responses, which is defined as follows:

$$\mathrm{PPE} = \left| \frac{|\hat{y}|_{\max} - |y|_{\max}}{|y|_{\max}} \right| \times 100\% \tag{10}$$

where $|y|_{\max}$ and $|\hat{y}|_{\max}$ represent the absolute peak values of the target and the predicted response sequences, respectively.

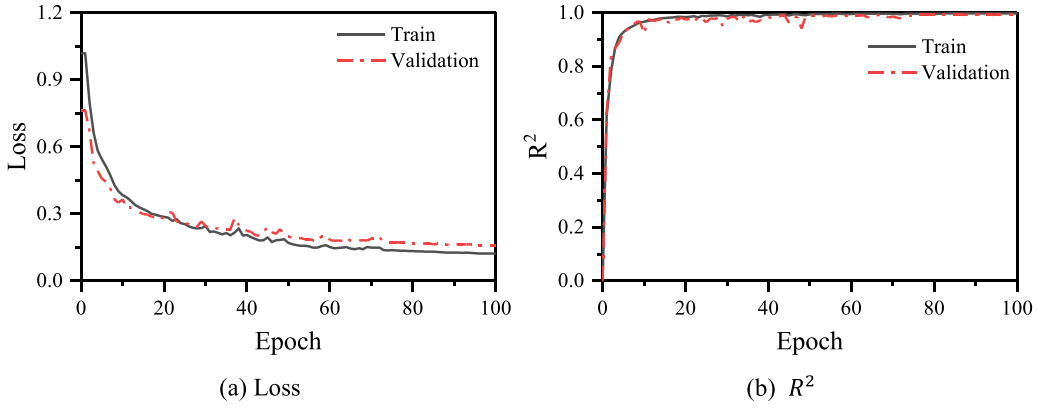
3.3. Test results analysis

Fig. 10 illustrates the convergence evolution of the loss function and the R^2 during the model training and validation process. Results indicate that the loss function decreases exponentially and stabilizes rapidly during the early stages of training. The R^2 increases and maintains stable in subsequent iterations. This validates that the proposed model possesses high optimization efficiency and stable mapping capabilities when processing highly nonlinear and non-stationary seismic response sequences.

Fig. 11 presents the comparison of moment response between the predicted and target time histories of the P3 pier under four typical cases in the test set, and the remaining comparison results are provided in Figure A2 of Appendix A. Overall, the predicted sequences maintain a high degree of consistency with the target responses in terms of global waveform morphology, phase synchronization and local transient peaks. To further evaluate the overall performance, statistical analysis based on 16 test seismic records demonstrates that the mean R^2 reaches 0.916, while the mean PPE is only 5.56% (Fig. 12), which further confirm the reliability of the model in full-duration fitting and extreme value characterization. Nevertheless, although the test set is independent at the ground-motion record level, it still shares the same structural model and general seismic scenario coverage as the training set. Therefore, the reported accuracy should not be interpreted as validation of the model for ground motions or damage states substantially outside the training distribution. Additionally, the R^2 value of one seismic record in the test set (0.78) is lower than the average, which is due to the low response amplitude for this specific case (approximately 900 kN×m). Under similar magnitude of absolute error, low-amplitude signals produce more intense fluctuations, leading to an increase in the sensitivity of the R^2 to deviations.

4. Hyperparameters sensitivity analysis

To evaluate the sensitivity of the proposed framework to key hyperparameters, the impacts of factors such as the number of SWT decomposition levels, encoder length, decoder length, hidden layer dimension and noise injection level on predictive performance are investigated.



(a) Loss (b) R^2

Fig. 10. Convergence curves of the loss function and R^2 during the training and validation phases.

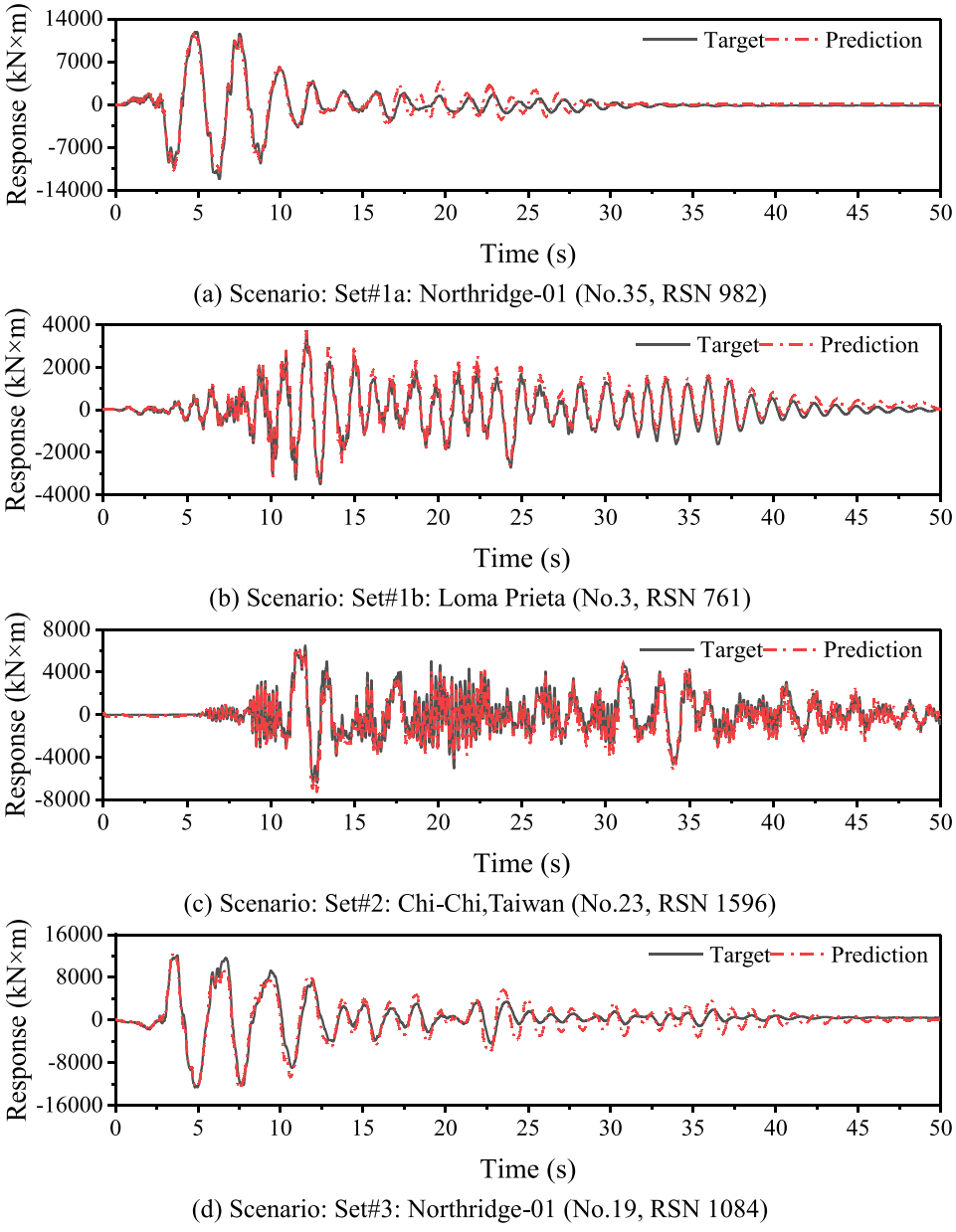


Fig. 11. Comparison of predicted and target response time histories under four typical test scenarios.

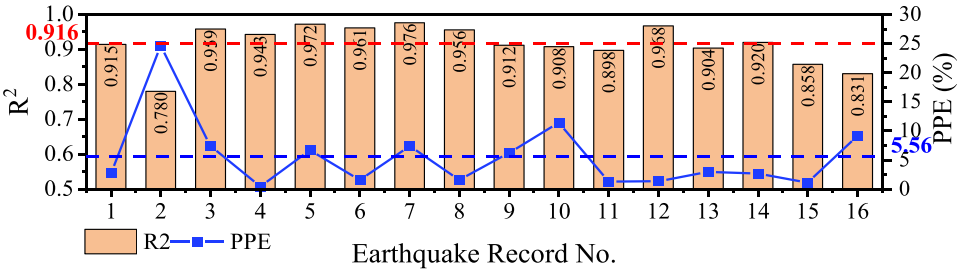


Fig. 12. R^2 and PPE across 16 ground motion scenarios in the test set.

4.1. Number of SWT decomposition levels

Fig. 13 illustrates the influence of decomposition levels N on the training evolution of the model while other hyperparameters remain constant. Results indicate that the training and validation losses for all configurations exhibit favorable convergence, with no significant signs of overfitting observed. The R^2 for $N = 7$ in Fig. 13(b) is zero because it fails to satisfy the signal length constraint ($2^7 = 128$) required for ISWT reconstruction. For $N < 5$, the loss curves exhibit significant sawtooth fluctuations, resulting from inadequate frequency band partitioning

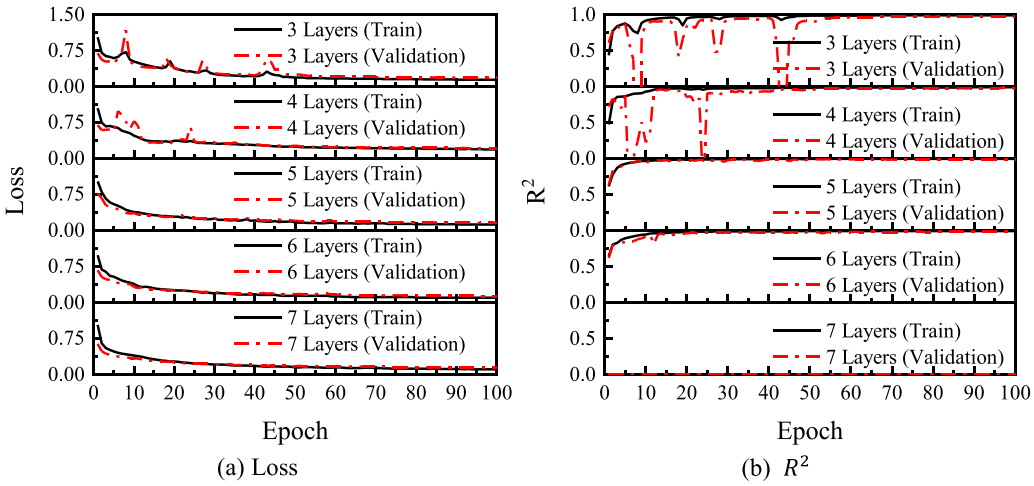


Fig. 13. Comparison of training convergence performance across different SWT decomposition levels.

degree. In contrast, for $\boldsymbol{N \geq 5}$, the loss decreases more smoothly, and the $\boldsymbol{R^2}$ is improved with faster early increase. This suggests that a finer multi-scale decomposition may help the model learn more stable mappings across different frequency bands.

Fig. 14 further compares the average comprehensive performance of whole sequence prediction on the test set under different decomposition levels, and the remaining comparison results of test set are documented in Tables A5-A6 of Appendix A. Results show that the model performance exhibits a trend of initially increasing and subsequently decreasing as $\boldsymbol{N}$ increases. At $\boldsymbol{N = 3}$ **or** $\boldsymbol{4}$, the insufficient scale representation leads limits in phase synchronization and local detail restoration. At $\boldsymbol{N = 5}$, the model achieves an optimal balance between scale resolution and learning difficulty, exhibiting the highest $\boldsymbol{R^2}$ and lowest PPE metrics. However, as $\boldsymbol{N}$ is further increased to 6 or 7, the linear expansion of input dimensions leads to insufficient representational capacity of the existing model.

4.2. Encoder sequence length

Fig. 15 shows the training evolution of the models under different L_{enc} while other hyperparameters remain constant. Overall, losses for all groups exhibit a stable downward trend, and R^2 is improved without obvious divergence between the training and validation sets, indicating consistent convergence and generalization capability within the investigated range.

Fig. 16 further compares the average comprehensive performance of whole sequence prediction on the test set under different L_{enc}, and the remaining comparison results of test set are documented in Tables A7-A8 of Appendix A. When $L_{\text{enc}} < 2^N$, the model fails with negative R^2 values, and predicted curves approximate a constant mean value

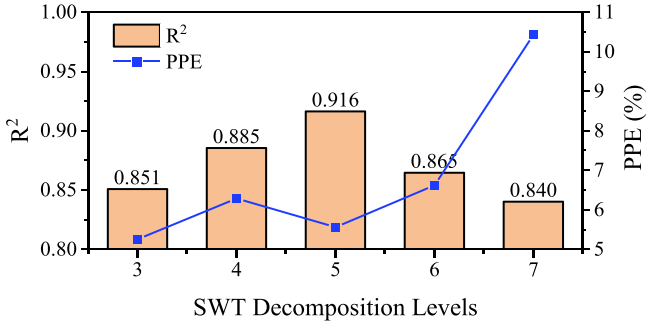


Fig. 14. Comparison of model performance across different decomposition levels $\boldsymbol{N}$.

(Fig. 17), resulting from the scale decomposition constraints of the SWT/ISWT. Excessively short input signals cause some low-frequency characteristic channels to become almost silent, thereby leading to numerical instability. Once L_{enc} satisfies the SWT length constraint, the prediction accuracy improves significantly as L_{enc} increases from 32 to 128. Longer windows includes more periods and decay information [57], thereby enhancing the identification ability of structural damage state. However, when L_{enc} is further increased to 256, performance slightly declines. Excessively long sequences may introduce redundant components weakly correlated with the prediction target, causing feature dilution in the attention mechanism. Furthermore, such redundancy increases the modeling complexity, leading to diminishing marginal returns or even slight performance degradation.

4.3. Decoder sequence length

Fig. 18 shows the train evolution of model under different decoder lengths L_{dec} while other hyperparameters remain constant. Results demonstrate that the losses across all investigated case exhibit a stable decreasing trend, and R^2 increases steadily with the training epochs, with no significant sign of overfitting observed. Similar to the encoder, when $L_{dec} < 2^N$, numerical collapse occurs due to the violation of the sequence length constraint of ISWT. As $L_{dec} > 64$, R^2 exhibit poor model performance stability with pronounced sawtooth oscillations. With a fixed model capacity, the decoder needs to simultaneously capture nonlinear behaviors over a longer time domain, significantly increasing the learning difficulty of autoregressive generation and thus reducing model stability.

Fig. 19 further compares the average comprehensive performance of whole sequence prediction on the test set under different L_{dec}, and the remaining comparison results of test set are documented in Tables A9-A10 of Appendix A. For $L_{dec} < 2^N$, the output contains significant high-frequency noise (Fig. 20), leading to negative R^2 values, which is caused by the sharp increase in the number of predictions windows under the slide window setting. To generate response sequences of equal length, the model needs to perform frequent block-level iterations and concatenation, and the phase and amplitude discontinuities at the sub-segment boundaries induce non-physical high-frequency oscillations. For $L_{dec} = 32$, the low R^2 is primarily due to the inability of short prediction spans to effectively suppress exposure errors (Fig. 21). When $L_{dec} > 64$, the model exhibits higher R^2 and lower PPE. This indicates that a longer L_{dec} can suppress intra-segment exposure error and reduce inter-segment error accumulation by decreasing the number of rolling iterations. Together with Fig. 18, although decoders with shorter lengths facilitate accelerated convergence and reduced training errors, such architectures

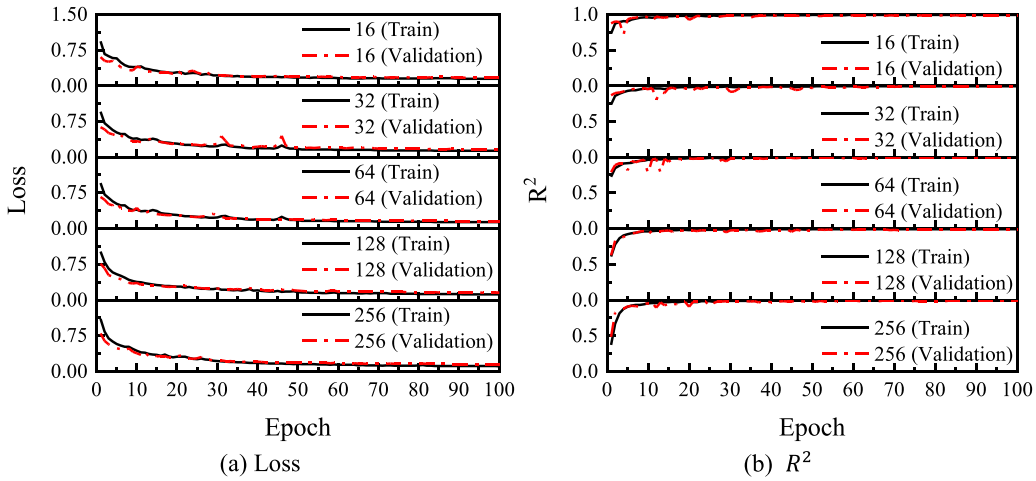


Fig. 15. Comparison of training convergence performance across different encoder lengths L_{enc}.

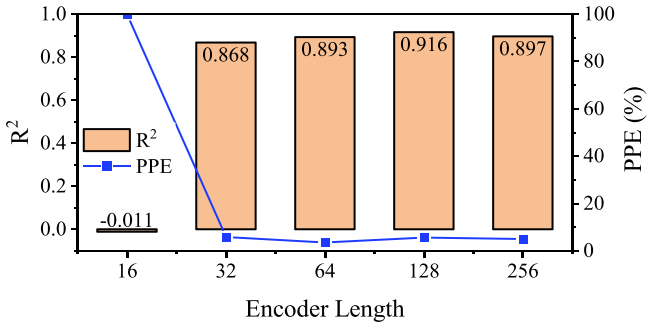


Fig. 16. Comparison of model performance across different encoder length L_{enc}.

may suffer from performance degradation in iterative, long-sequence inference scenarios.

4.4. Hidden layer dimension

Fig. 22 illustrates the influence of hidden layer dimension d_{model} on the training evolution of the model while other hyperparameters remain constant. The losses across all configurations exhibit a sable decreasing trend, with no significant sign of overfitting observed. R^2 increases rapidly during the initial stages and then gradually plateaus when the d_{model} exceeds 64. Minor fluctuations are observed in the loss and R^2 curves during the initial stages when the d_{model} exceeds 128, which is because parameter redundancy introduced by the high-dimensional latent space leads to instability in gradient search during the early stages of optimization.

Fig. 23 further compares the average comprehensive performance of whole sequence prediction on the test set under different d_{model}, and the remaining comparison results of test set are documented in Tables A11-A12 of Appendix A. Results demonstrate that when the d_{model} is 32, the model exhibits significant numerical failure, with negative R^2 value and high PPE. The low d_{model} leads to insufficient latent space capacity, hindering the efficient encoding of multi-scale coefficients. As the d_{model} increases to 64 and 128, the model's prediction accuracy reaches the peak ($R^2 \approx 0.92$, with a significantly reduced PPE), achieving an optimal balance between multi-scale feature representation and model capacity. However, upon further increasing the dimension to 256 and 512, a slight decrease in the test set R^2 is observed. Under conditions of constant data scale and regularization intensity, an excessively large latent space capacity leads to overfitting of the noise in training set, thereby undermining generalization stability across different earthquake samples.

4.5. Noise injection level

Fig. 24 illustrates the influence of noise intensity configurations on the training evolution of the model while other hyperparameters remain constant. In general, losses under all noise levels exhibit a robust downward trend, which indicates that the introduction of moderate perturbations does not interfere with the convergence performance of the model. As noise intensity increases, losses and R^2 during the early training stage exhibit more pronounced fluctuations, resulting from the corruption of the ground-truth signals by excessive random perturbations, which increases the uncertainty of gradient optimization in the initial stages.

Fig. 25 further compares the average comprehensive performance of whole sequence prediction on the test set under different noise levels, and the remaining comparison results of test set are provided in Tables A13-A14 of Appendix A. In the absence of noise injection, the R^2 remains at a relatively low level due to slight drift caused by exposure error. As the noise level increases to 0.05 and 0.10, R^2 improves

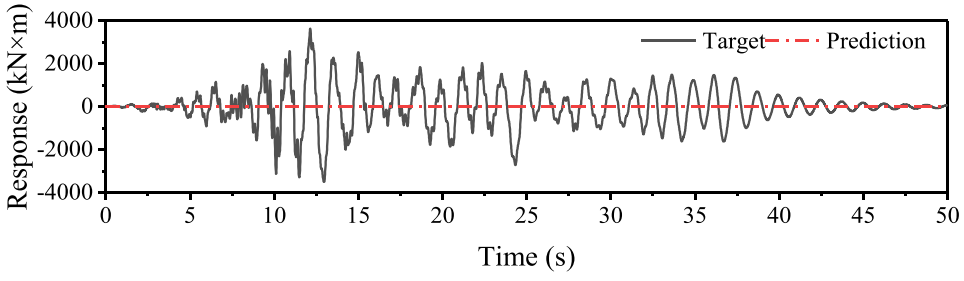


Fig. 17. Comparison of predicted and target response time histories with an encoder length of $L_{enc} = 16$ (Scenario: Set#1b: Loma Prieta (No.3, RSN 761)).

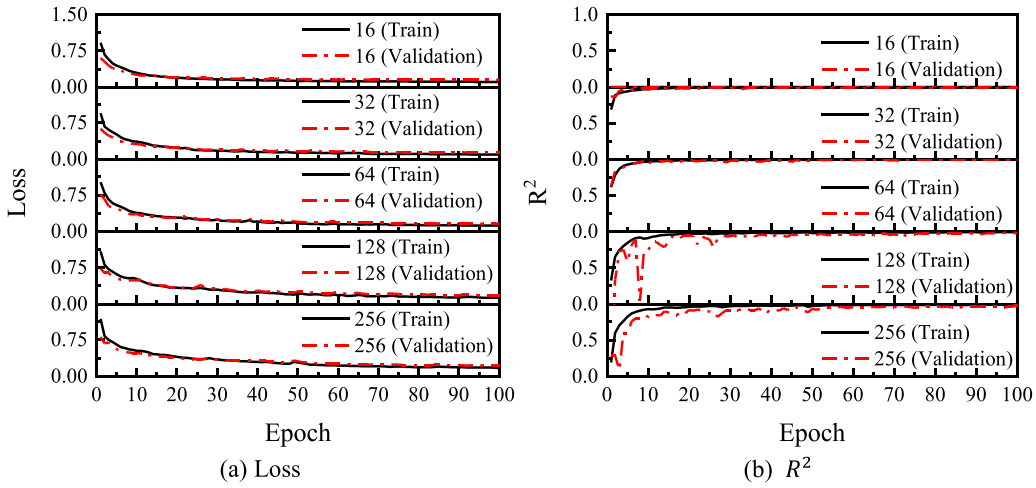


Fig. 18. Comparison of training convergence performance across different decoder lengths L_{dec}.

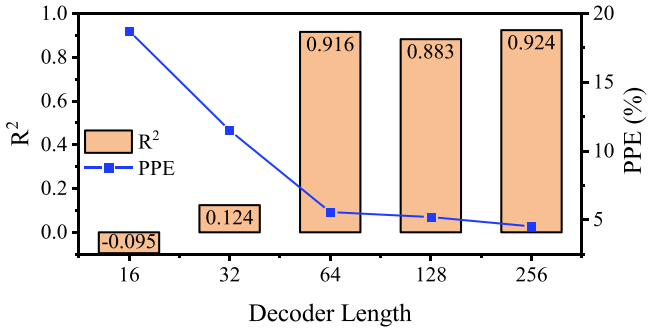


Fig. 19. Comparison of model performance across different decoder length L_{dec}.

significantly and reaches a peak of approximately 0.925 at a noise level of 0.10, which suggests that moderate noise injection induces more robust state update criteria. However, when the noise level is further increased to 0.15 and 0.20, a slight decline in R^2 is observed, which reflects that excessive perturbations reduce the effective signal-to-noise ratio in the feature space and lead to the loss of critical time-frequency details.

5. Ablation study

To evaluate the necessity of the key components within the proposed framework, ablation studies are conducted in this section focusing on critical dimensions, including the necessity of multi-scale decomposition, wavelet channel redundancy processing strategies and loss function design.

5.1. Multi-scale feature decomposition

This ablation study aims to compare the performance disparities between the multi-scale modeling strategy and the traditional time-domain modeling approach. The control group uses a traditional time-domain Transformer prediction model (hereafter referred to as the Transformer model), which maintains a network structure identical to

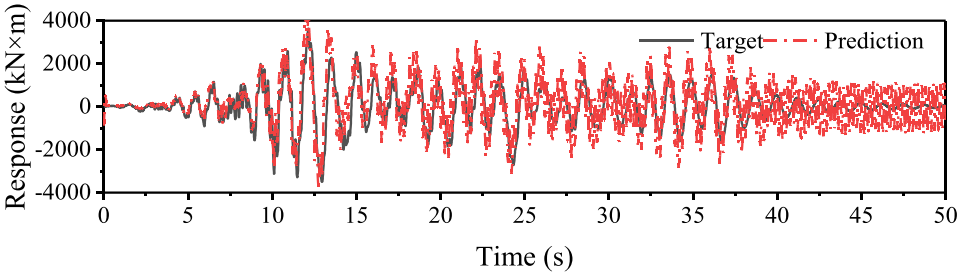


Fig. 20. Comparison of predicted and target response time histories with a decoder length of $L_{dec} = 16$ (Scenario: Set#1b: Loma Prieta (No.3, RSN 761)).

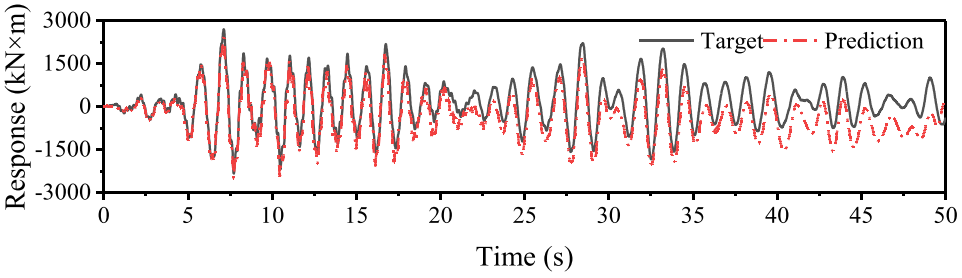


Fig. 21. Comparison of predicted and target response time histories with a decoder length of $L_{dec} = 32$ (Scenario: Set#1b: El Alamo (No.15, RSN 22)).

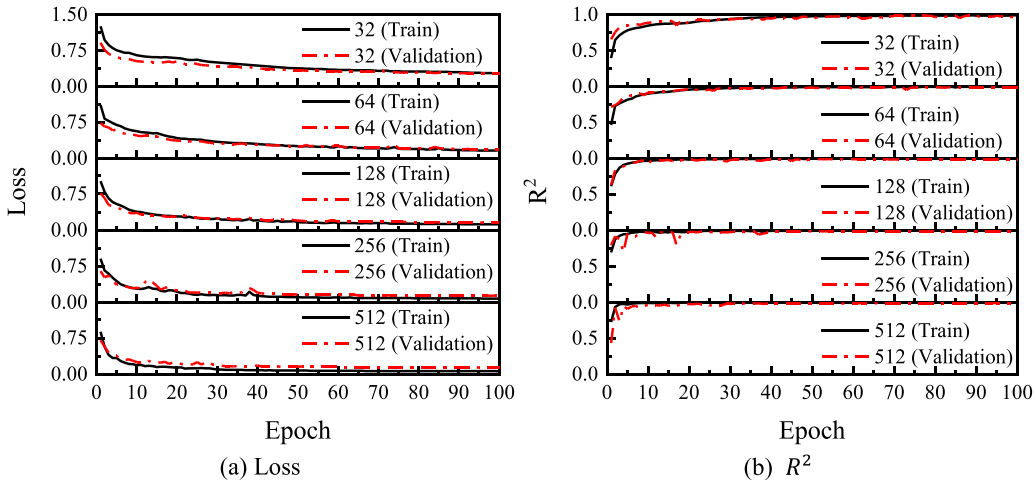


Fig. 22. Comparison of training convergence performance across different hidden layer dimension.

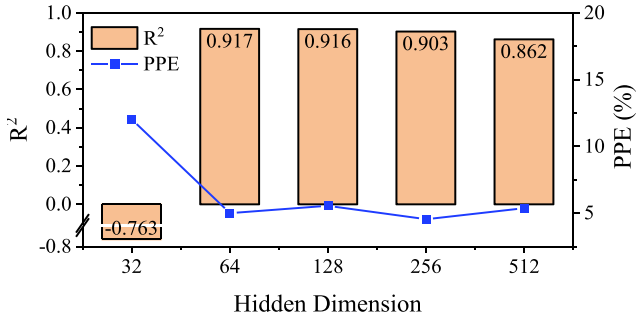


Fig. 23. Comparison of model performance across different hidden layer dimension.

that of the proposed model (hereafter referred to as the SWT model). The SWT decomposition and ISWT reconstruction modules are removed from the control group to directly establish the mapping relationship between the seismic excitation and the structural response time histories within the time domain.

Compare Fig. 10 and Fig. 26, the Transformer model exhibits significant random fluctuations and convergence instability during the initial optimization stage. By contrast, the SWT model demonstrates better stability and robustness throughout the entire training epoch. The multi-resolution analysis of the SWT decouples non-stationary excitations and responses into different frequency-band subspaces, which facilitates the effective separation of low-frequency trends and high-frequency transients at the representation level, thereby enhancing the network's capacity to characterize complex structural dynamic evolution processes.

Fig. 27 presents the comprehensive performance results of the Transformer model for the whole prediction task on the test set, yielding mean R^2 and PPE values of 0.849 and 6.09%, respectively. A further comparison between Fig. 12 and Fig. 27 indicates that the mean values of all evaluation metrics for the Transformer model are inferior to those

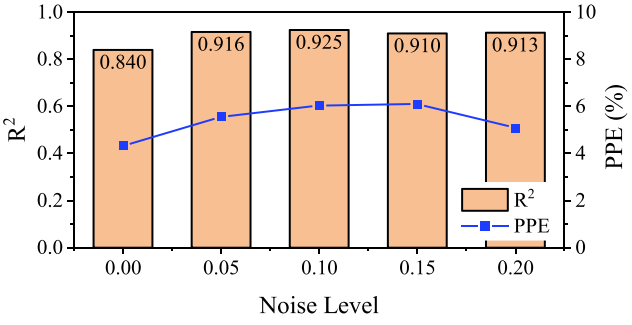


Fig. 25. Comparison of model performance across different noise level.

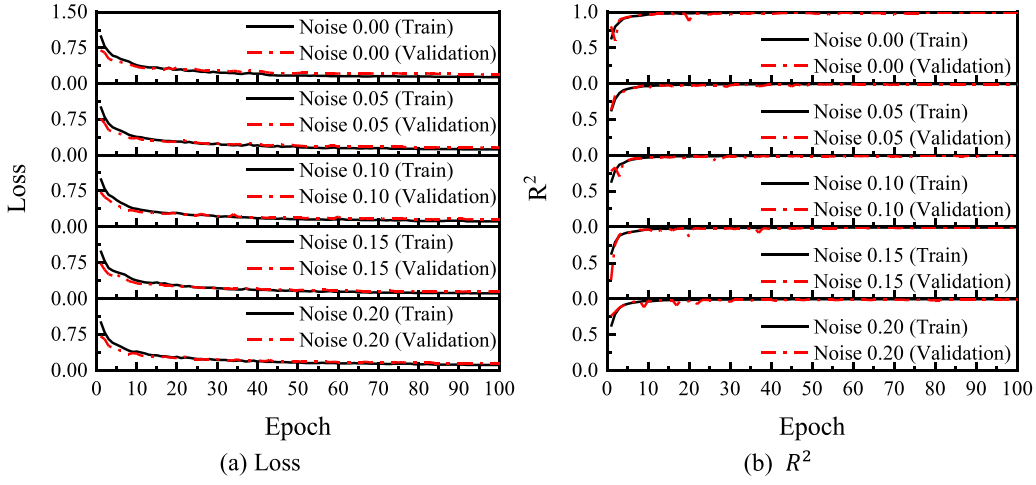


Fig. 24. Comparison of training convergence performance across different noise level.

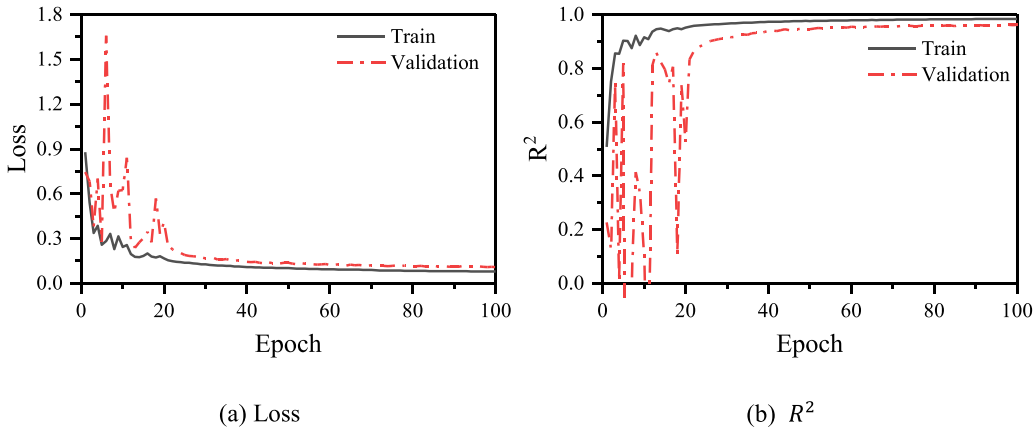


Fig. 26. Comparison of training convergence stability between Transformer model and SWT model.

of the SWT model, verifying the key role of multi-scale feature enhancement in improving prediction performance. Furthermore, the performance of the Transformer model exhibits significant dispersity across different samples, particularly in terms of the R^2 metric. In processing high-frequency transient responses (Fig. 28), the SWT model demonstrates higher waveform reconstruction fidelity, whereas the Transformer model exhibits an insufficient capability to represent high-frequency information and exhibits slight drift. In closed-loop recursions lacking frequency-band constraints, minor single-step deviations tend to accumulate errors, causing the predicted results to gradually deviate from the ground-truth values. In contrast, the SWT model effectively suppresses error accumulation and amplification through multi-channel representation and cross-scale supervisory constraints, thereby ensuring the stability of long-sequence predictions.

5.2. Scale redundancy representation

To explore the necessity of feature redundancy in multi-scale decomposition (Eqs. 2 and 3), a channel trim ablation study is conducted. The prediction performance of the full-channel input strategy (SWT model) is compared with that of the approximation channel trim strategy (SWT_Trim model). The latter approach retains only the deepest cA_N and cD_j at each level as input to examine the impact of scale redundancy on model convergence and prediction stability.

Compare Fig. 10 and Fig. 29, although the SWT_Trim model can achieve convergence, it exhibits instability during the early stages of training. Furthermore, the validation set is generally inferior to that of the full-channel SWT model, indicating that the removal of shallow approximation channels compromises the optimization stability and generalization performance of the model. The SWT_Trim model can also fit dynamic features in short decoder segments, which suggests that the trim strategy does not entirely obstruct the model's ability to capture local dynamic behaviors.

Fig. 30 presents the comprehensive performance results of the SWT_Trim model for the whole prediction task on the test set. Results indicate that the inference performance of the SWT_Trim model performs poor, with an average R^2 of only −0.05 and obvious amplitude deviation and phase misalignment (Fig. 31). Compare Fig. 12 and Fig. 30, both the R^2 and PPE of the model deteriorate significantly after the approximation channels are trimmed, thereby revealing the necessity of redundant channels representation. Although the shallow cA_j overlaps with the deep cA_N, it provides a critical low-frequency constraint with high temporal resolution during the autoregressive recursive process. Under conditions of scale imbalance, where the low-frequency baselines are lacking while multi-level high-frequency details are retained, the model tends to overfit high-frequency perturbations to minimize instantaneous residuals. Due to the absence of the low-frequency constraint, local prediction deviations rapidly transform into global baseline drift and phase disorder during closed-loop recursion, causing the predicted evolution to deviate from the actual dynamic responses. Although channel trimming reduces feature dimension, it disrupts the structured redundancy and multi-scale fault-tolerance mechanisms. Therefore, for tasks involving highly non-stationary autoregressive inference, maintaining multi-scale redundant representation is an effective way of suppressing error accumulation and ensuring long-sequence prediction stability.

5.3. Selection of loss functions

Given that the loss function directly defines the update direction of model, this section presents a systematic ablation study to compare the impacts of MSE, MAE and the hybrid loss strategy on training dynamics and prediction performance.

As illustrated in Fig. 32(a), the model under MSE constraints exhibits a larger loss value during the initial training phase but end with a higher convergence level. Fig. 32(b) further confirms that the MSE possesses higher convergence efficiency. Compare Fig. 10 and Fig. 32, the convergence evolution and steady-state level of the hybrid loss strategy lie between those of MSE and MAE, reflecting a synthesis of the constraint effects of both functions.

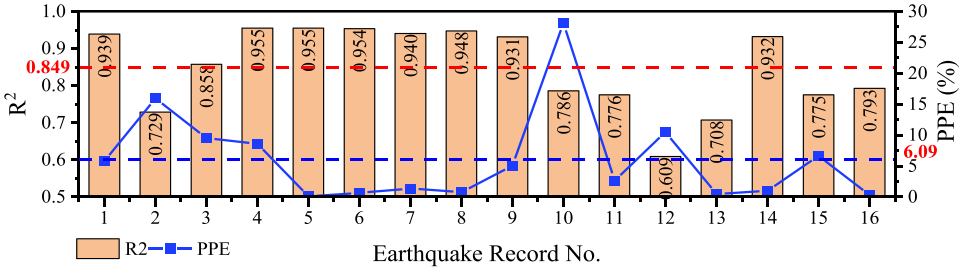


Fig. 27. Results of R^2 and PPE across 16 ground motion in the test set with Transformer model.

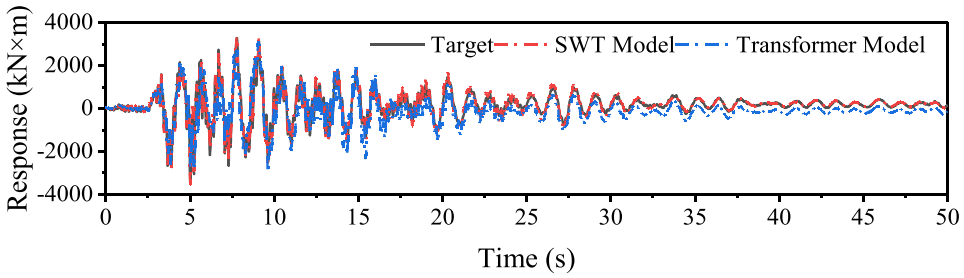


Fig. 28. Comparison of predicted and target response time histories across different autoregressive models (Scenario: Set#2: Northridge-01 (No.27, RSN 1078)).

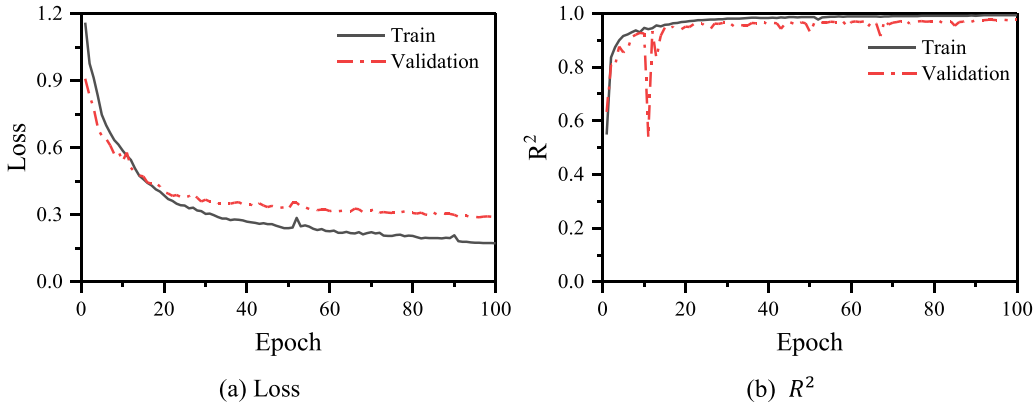


Fig. 29. Comparison of training convergence stability between SWT_Trim model and SWT model.

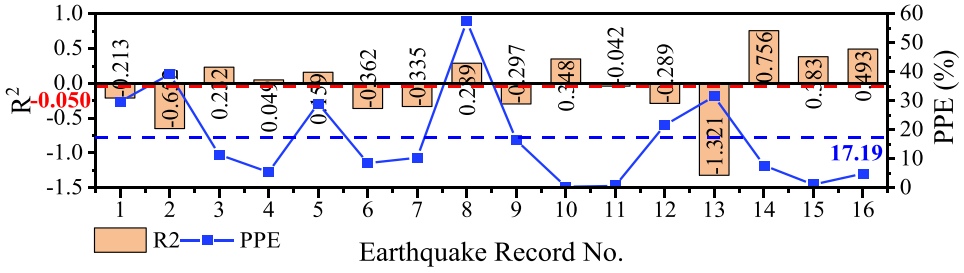


Fig. 30. Results of R^2 and PPE across 16 ground motion in the test set with SWT_Trim model.

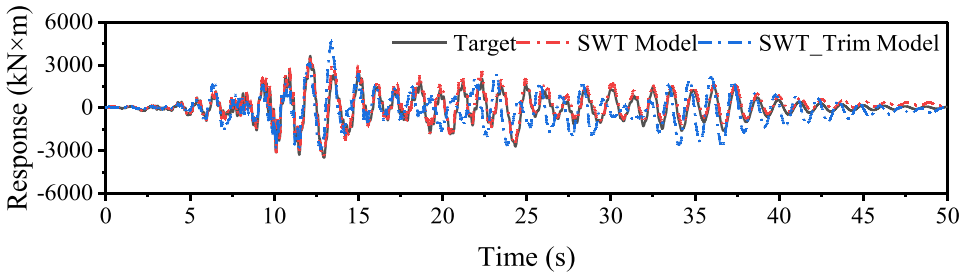


Fig. 31. Comparison of predicted and target response time histories across different autoregressive models (Scenario: Set#1b: Loma Prieta (No.3, RSN 761)).

Fig. 33 provides the comprehensive performance of the models under different loss function for the whole prediction task on the test set. When trained under the MSE loss constraint, the model achieved an average R^2 of 0.882 and a PPE of 4.65%. Conversely, utilizing the MAE loss constraint resulted in an average R^2 and PPE of 0.897 and 6.07%. These results reveal a divergence in performance preference under different loss function constraints. Specifically, the MAE more effectively enhances the overall fitting consistency of the waveform, resulting in a higher R^2. In contrast, the MSE renders more sensitive to signal extreme amplitudes, leading to superior performance in PPE. The hybrid loss strategy adopted in this study couples the extremal constraints of the MSE with the morphological fidelity constraints of the MAE, which significantly reduces prediction deviations at critical peaks while simultaneously maintaining global trend fitting.

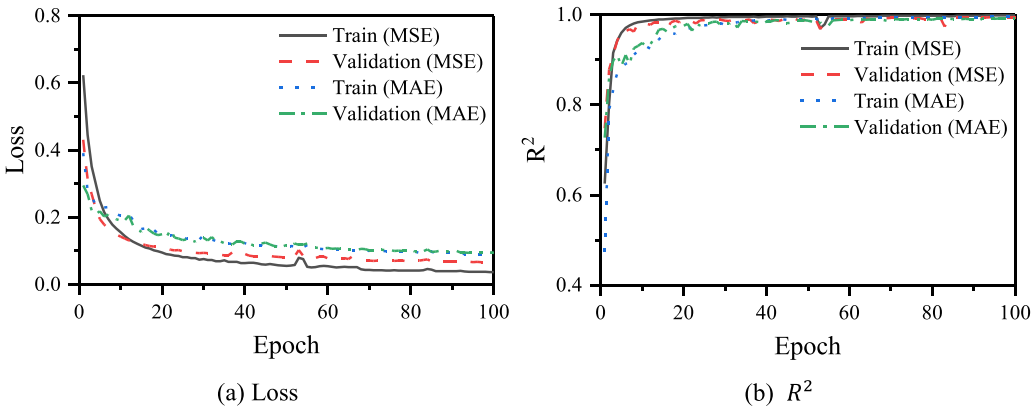


Fig. 32. Comparison of training convergence stability between different loss function constraints.

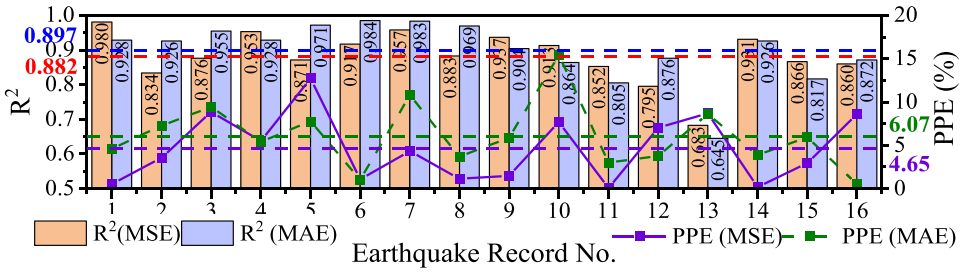


Fig. 33. Results of R^2 and PPE across 16 ground motion in the test set with different loss function constraints.

6. Discussion

The present study focuses on the bending moment at the base of pier because it is directly related to pier damage assessment and contains both global modal components and local transient features associated with nonlinear behavior. This makes it a suitable response quantity for examining the multi-scale representation capability of the proposed framework. However, the results obtained for bending moment cannot establish equivalent performance for other engineering demand parameters, such as displacement, inter-story drift, shear force and acceleration. In principle, the proposed SWT-Transformer framework can be extended to scalar or multichannel structural responses because SWT and inverse SWT are applicable to general time series and the Transformer output dimension can be adjusted accordingly. Nevertheless, different response quantities have different frequency characteristics and may require specific model settings. Low-frequency responses, such as displacement and inter-story drift, may require longer observation windows and stronger drift control, whereas high-frequency responses, such as acceleration and local force responses, may require higher sampling rates, different decomposition levels and response-specific noise-injection parameters. Therefore, the present results demonstrate the feasibility of the framework for nonlinear structural seismic response prediction, but do not establish universal applicability or identical accuracy for all engineering response quantities. Further validation using multiple response types and measured structural data is required.

All input-output samples used in this study were generated through nonlinear finite-element time-history analyses using OpenSees, which provides synchronized ground-motion and structural-response records under controlled structural and loading conditions. It also enables different nonlinear response states to be represented systematically and provides a sufficiently large dataset for model training, validation, and testing. However, the numerical simulations do not fully represent the uncertainties associated with field measurements. The adopted finite-element model may involve simplifications in structural properties, soil-structure interaction and boundary conditions. In addition, actual measurements may be affected by construction and maintenance variability, component degradation, baseline drift, missing channels, sampling errors and synchronization errors. These factors are not fully reproduced by the simulated data.

Therefore, the results reported in this study demonstrate the performance of the proposed framework for unseen ground-motion records within the response distribution represented by the adopted OpenSees model and ground-motion database. Further validation using instrumented bridges or buildings subjected to recorded earthquakes is required. Future studies should also investigate measurement-noise modeling, domain adaptation and fine-tuning with a limited number of measured response records to improve the transferability of the proposed framework to practical engineering applications.

7. Conclusions

An autoregressive structural seismic response prediction framework is proposed by integrating the SWT and the Transformer architecture to achieve high-precision prediction of non-stationary structural responses. The proposed framework, through the autoregressive paradigm, enables short-term rolling prediction of structural seismic response signals. Compared to nonlinear time-history analysis, it offers a faster inference speed, especially for complex structures. Based on the seismic responses of a typical continuous girder bridge, alongside systematic parameter sensitivity analyses and ablation studies, the following conclusions are drawn:

- The proposed framework demonstrates superior performance, yielding a mean R^2 of 0.916 and a PPE of 5.56% on the test set, while maintaining high fidelity in reconstructing complex dynamic evolution waveforms and high reliability in capturing key extreme responses.
- Compared with the time-domain Transformer baseline under the same prediction task, the SWT-based coefficient representation improves high-frequency detail reconstruction and stabilizes training

convergence by separating low-frequency trends and high-frequency components.

- The inherent redundant channels in the multi-scale decomposition provide high-resolution low-frequency constrains, preventing minor single-step deviations from evolving into systematic baseline drifts, thereby ensuring the stability of autoregressive predictions.
- The channel-wise noise injection and self-conditioning training strategies during training stage effectively mitigates exposure bias in the adopted rolling prediction setting, thereby enhancing the stability of long-sequence autoregressive predictions.
- The number of SWT decomposition levels, encoder/decoder window lengths, hidden layer dimensions and noise levels demonstrate significant influence on model performance. The selection of these hyperparameters should balance the sufficiency of feature representation and the robustness of model generalization, avoiding performance degradation caused by parameter redundancy.

CRediT authorship contribution statement

Pan Liu: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Jiajun Du:** Visualization, Data curation. **Quincy Ma:** Writing – review & editing, Supervision. **Jianzhong Li:** Supervision. **Zhongguo Guan:** Resources. **Hongya Qu:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

The corresponding Tables and Figures mentioned in the main text are provided in supplementary data.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.istruc.2026.113507.

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